Tasks and Parallelism

Running work in the background and getting results back: futures, thread pools, work stealing, the C++20 coordination tools and parallel algorithms. Builds on Multithreading.

Futures and async

To **run a function in the background and get its result later**, use a `std::future<T>`.

```
#include <future>

int slowSquare(int n) { std::this_thread::sleep_for(std::chrono::seconds(1));
return n * n; }

std::future<int> f = std::async(std::launch::async, slowSquare, 7);   //
starts on another thread
doOtherWork();                                                          //
main keeps working meanwhile
int result = f.get();                                                   //
wait if needed, then get 49
```

- `f.get()` blocks until ready (once only); `wait()` and `wait_for()` do not take the value.
- **Exceptions travel**: a throwing task rethrows from `f.get()`.
- **Prefer tasks to raw threads**: `std::thread` has no result channel and terminates on exceptions.
- Pass `std::launch::async`; the default may *defer* the call to `get()`.
- An `async` future **blocks in its destructor**, so a discarded one makes the call effectively synchronous.

```cpp
// std::promise / std::future: a one-shot channel between two threads you manage yourself
std::promise<int> p;
std::future<int> f = p.get_future();
std::thread t([&p] { p.set_value(42); });    // the worker fulfils the promise (or p.set_exception( ... ))
std::cout << f.get();                        // 42
t.join();
```

`std::packaged_task` routes a function's result into a future (how pools return results).

`promise`, `shared_future` and `packaged_task`

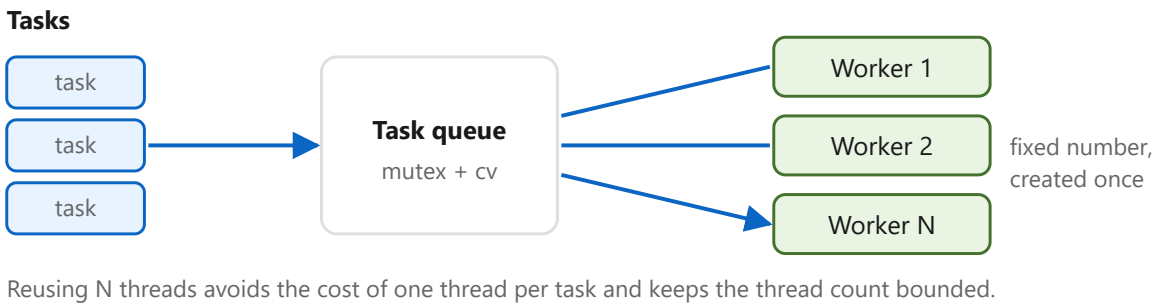
TYPE	ROLE
`std::promise<T>`	the **writing** end: `set_value` or `set_exception`, once
`std::future<T>`	the **reading** end: `get()` once; move-only
`std::shared_future<T>`	a **copyable** read end: several threads can `get()` or `wait()` on the same result
`std::packaged_task<R(Args)>`	wraps a callable and routes its result (or exception) into a future: how a pool returns results

```cpp
// promise<void>: a one-shot "go" signal for several waiting threads
std::promise<void> go;
std::shared_future<void> start = go.get_future().share();     // copyable, so every worker gets a handle
for (int i = 0; i < 4; ++i) workers.emplace_back([start] { start.wait(); run(); });
go.set_value();                                               // all four are released together

// packaged_task: the pool wraps a function and hands back the future
std::packaged_task<int(int)> task([](int n) { return n * n; });
std::future<int> f = task.get_future();
std::thread t(std::move(task), 7);                            // runs on any thread
int r = f.get();                                              // 49
t.join();
```

- Compiled: four workers waiting on a shared `promise<void>` had started 0 before `set_value()` and 4 after.
- A `promise` destroyed without a value gives the future a `broken_promise` exception instead of hanging forever.
- For a repeated signal use a condition variable; a promise is **one-shot**.
- A future is a *one-way, one-time* channel. If you need many messages, use a queue.

Thread pools and producer/consumer



Reusing N threads avoids the cost of one thread per task and keeps the thread count bounded.

Thread pool: many tasks, few long-lived workers.

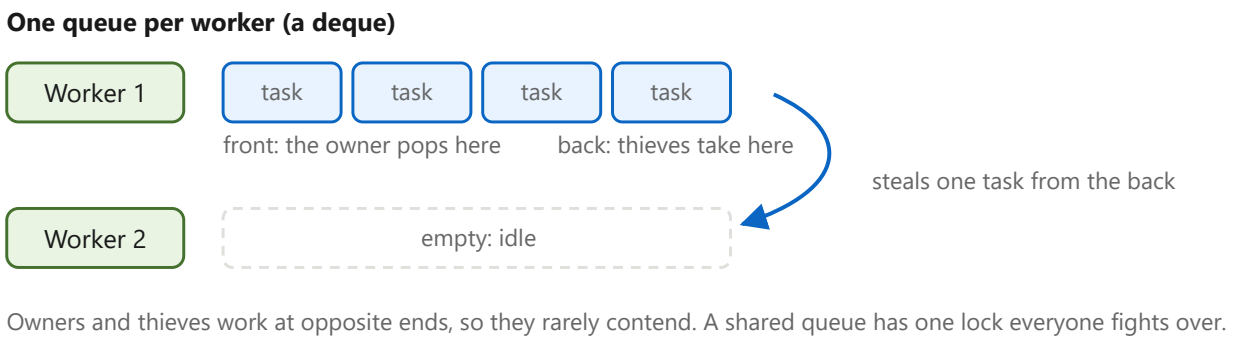
A thread per task is expensive and thousands overload the scheduler. A **thread pool** starts N workers once; tasks go in a queue (**producer/consumer**).

```
class ThreadPool {
    std::vector<std::thread> workers_;
    std::queue<std::function<void()>> tasks_;
    std::mutex m_;
    std::condition_variable cv_;
    bool stop_ = false;
public:
    explicit ThreadPool(std::size_t n) {
        for (std::size_t i = 0; i < n; ++i)
            workers_.emplace_back([this] {
                for (;;) {
                    std::function<void()> task;
                    {
                        std::unique_lock<std::mutex> lk(m_);
                        cv_.wait(lk, [this] { return stop_ ||
!tasks_.empty(); });
                        if (stop_ && tasks_.empty()) return;     // finish
remaining tasks, then exit
                        task = std::move(tasks_.front());
                        tasks_.pop();
                    }                                            // lock
released BEFORE running the task
                    task();
                }
            });
    }
    void submit(std::function<void()> f) {
        { std::lock_guard<std::mutex> lk(m_); tasks_.push(std::move(f)); }
        cv_.notify_one();
    }
    ~ThreadPool() {
        { std::lock_guard<std::mutex> lk(m_); stop_ = true; }    // request
shutdown under the lock
        cv_.notify_all();                                        // wake
every sleeping worker
        for (auto& w : workers_) w.join();                       // wait for
them to finish
    }
};

ThreadPool pool(std::thread::hardware_concurrency());
pool.submit([] { std::cout << "task runs on some worker\n"; });
```

It combines everything: mutex, condition variable, stop flag, RAII shutdown. The task runs **outside** the lock, or only one could run at a time.

Work stealing and sizing a pool



Work stealing: local deques keep the common case lock-light, stealing balances the load.

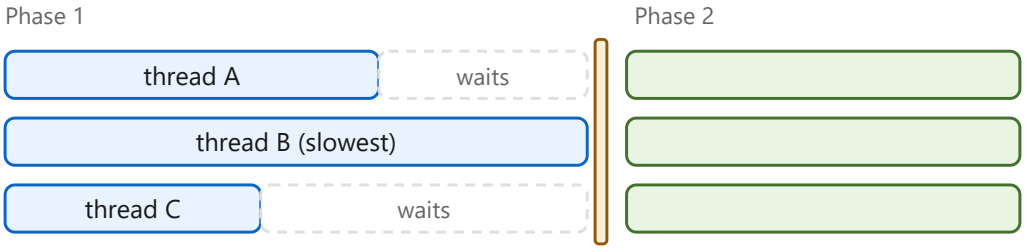
- **Work stealing** is how production pools (TBB, Java's fork/join) balance load: each worker pops its own deque and an idle worker **steals** from another's back.
- It suits **recursive** tasks that spawn subtasks: a spawned task goes on the spawner's own deque and stays hot in its cache.
- A simple pool with one shared queue (the previous section) is fine until that queue's lock becomes the bottleneck. Know that stealing is the next step; do not claim to have written one.

WORK	WORKERS
CPU-bound	about `hardware_concurrency()`; more is oversubscription and context switches
I/O-bound or blocking	more than the core count, because threads spend their time waiting
Mixed	separate pools, so slow blocking tasks cannot starve the fast ones

> **POOL DEADLOCK** A task that submits a subtask to the same pool and **waits** for it can block every worker: all N workers wait for subtasks nobody is free to run. Do not block a pool thread on work that needs the same pool.

`latch`, `barrier`, `semaphore` and `stop_token` (C++20)

TOOL	IDEA	REUSABLE?
`std::latch(n)`	wait until a counter has been counted down *n* times ("N things are done")	no: one shot
`std::barrier(n)`	all *n* threads wait for each other at the end of each phase, then all go on	**yes**: phase after phase
`std::counting_semaphore<N>`	at most *k* threads in a section (a permit counter)	yes
`std::binary_semaphore`	a semaphore with one permit: signal from one thread to another	yes



The barrier (vertical bar): the last arrival releases everyone, so phase 2 starts together.
Compiled: every thread saw all three phase-1 results before it began phase 2.

A barrier separates phases of a parallel computation.

```
std::latch done(3);                                    // three workers will
count down
for (int i = 0; i < 3; ++i) workers.emplace_back([&] { work();
done.count_down(); });
done.wait();                                           // returns when the
count reaches 0

std::barrier sync(3);                                  // three threads, many
phases
auto body = [&] { phase1(); sync.arrive_and_wait(); phase2(); };

std::counting_semaphore<4> slots(2);                   // at most two threads
at once
void limited() { slots.acquire(); use(); slots.release(); }   // compiled:
peak was 2 with six threads
```

Cooperative cancellation: `jthread` and `stop_token`

```
std::jthread worker([](std::stop_token st) {
    while (!st.stop_requested()) { doWork(); }       // check the token
regularly
});
// worker.request_stop();  or just let the destructor do it: request_stop()
then join()
```

There is no safe way to *kill* a thread; you ask it to stop and it finishes its loop. `std::stop_token` is the standard channel for that request, and `std::jthread` joins in its destructor, which removes the "forgot to join" crash.

Parallel algorithms (C++17)

Most standard algorithms accept an **execution policy** as a first argument. The library decides how to split the work across threads.

```
#include <execution>

std::sort(std::execution::par, v.begin(), v.end());
long long sum = std::reduce(std::execution::par, v.begin(), v.end(), 0LL);
// compiled: 500500 for 1..1000
std::transform(std::execution::par_unseq, a.begin(), a.end(), out.begin(),
f);
```

POLICY	MEANING
`seq`	sequential, as without a policy
`par`	may run on several threads
`par_unseq`	several threads and vectorised (SIMD): the function must not lock or allocate

- **Your function object must be thread-safe**: no unsynchronised writes to shared data. A data race inside a parallel algorithm is still undefined behaviour.
- `std::reduce` (not `accumulate`) is the parallel fold: it may **reassociate** the operation, so the operation must be associative (floating-point sums may differ slightly).
- It only pays for **big inputs and heavy work**; for small ranges the thread overhead dominates.
- With g++ you usually need Intel TBB installed and `-ltbb` for real parallelism; without a parallel backend the same code runs serially (as in the compiled example here).
- Exceptions escaping the callback call `std::terminate`.

> CHOOSE Parallel algorithms for data parallelism over a container; a pool or `std::async` for independent tasks; a queue and a pinned thread for a latency-critical pipeline (Performance & Memory).

Further reading by section

Where each section of this lesson is explained in more depth. Pages are PDF viewer pages (the number your PDF reader shows).

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
Futures and async	Effective Modern C++	Item 35, Prefer task-based programming to thread-based	259 to 263
	Effective Modern C++	Item 36, Specify std::launch::async if asynchronicity is essential	263 to 268
	Effective Modern C++	Item 38, Be aware of varying thread handle destructor behavior	276 to 280
	C++ Move Semantics	Chapter 15 Move semantics in standard library types	229 to 250
	C++ Concurrency in Action	4.2 Waiting for one-off events with futures (std::async, packaged_task, promise, shared_future)	104 to 115
	C++ Concurrency in Action	4.3 Waiting with a time limit	116 to 121
promise, shared_future and packaged_task	C++ Concurrency in Action	4.2 Waiting for one-off events with futures (std::async, packaged_task, promise, shared_future)	104 to 115
Thread pools and producer/consumer	C++ Concurrency in Action	9.1 Thread pools (a simple pool, waiting for tasks, work stealing in 9.1.5)	324 to 337
Work stealing and sizing a pool	C++ Concurrency in Action	2.4 Choosing the number of threads at runtime	54 to 56
	C++ Concurrency in Action	8.1 Techniques for dividing work between threads	275 to 282
	C++ Concurrency in Action	9.1 Thread pools (a simple pool, waiting for tasks, work stealing in 9.1.5)	324 to 337
latch, barrier, semaphore and stop_token (C++20)	C++ Concurrency in Action	4.4.7 to 4.4.10 Latches and barriers (the Concurrency TS versions of C++20 std::latch and std::barrier)	141 to 147
	C++ Concurrency in Action	9.2 Interrupting threads (the idea behind std::stop_token)	338 to 349
Parallel algorithms	C++ Concurrency in Action	10.1 to 10.3 Parallel algorithms: execution policies and the parallel standard algorithms	350 to 362